

GPU (Unified Memory)

MIG Instance A (12 SMs)

MIG Instance B (6 SMs)

reads and writes in place

Serving Processes (One Address Space Each)

Publisher

Weights KV Cache

model prefix tail

①

Agent 1

Weights KV Cache

model prefix tail

Agent 2

Weights KV Cache

model prefix tail

②

③

Agent 3

Weights KV Cache

model prefix tail

Host Page Tables: one per process, walked by the GPU through the SMMU

4 mappings, 1 copy

4 mappings, 1 copy

1 mapping each

Model File

one copy, 2 MiB pages

Prefix Cache File (tmpfs)

one copy, immutable rows

names the rows

State File: metadata only



Private Tails

*anonymous memory,
grows with the rows in use*

① publish

② attach

③ grow the tail

— read-only mapping, shared pages

— writable mapping, pages of one process

Host Memory (One DRAM)